



Project Plan

Smart Fridge AI

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Section 1. Project Overview

1.1 Project Description

The Smart Fridge Management System is an AI-powered application designed to help users efficiently organize their food, track expiration dates, and receive personalized recipe recommendations based on available ingredients. The project aims to reduce food waste, enhance meal planning, and promote healthier eating habits through intelligent automation and data analysis.

1.2 Project Scope

1.2.1 project includes

Project Includes
IoT-Enabled Food Inventory Management - Automated tracking of food items using weight sensors and computer vision-based recognition. - Barcode scanning for quick and easy product registration.
Expiration Date Tracking & Alerts - Smart notifications to inform users about expiring items. - Categorization of food based on expiration status (e.g., fresh, expiring soon, expired).
User Health & Preference Customization - Tailored meal suggestions for users with specific dietary needs (e.g., vegan, gluten-free). - Nutritional insights and calorie tracking.
Real-Time IoT Integration - Weight sensors automatically update inventory when an item is added or removed. - Camera-based food recognition detects and updates stock levels. - Barcode scanner integration ensures accurate product entry. - Seamless cloud synchronization to keep inventory updated in real time.
Smart Recipe Suggestions - AI-based personalized meal recommendations using available ingredients. - Customizable recipes based on dietary preferences and restrictions.
Data Analysis & Insights - Reports on food consumption patterns and waste reduction strategies. - Smart shopping suggestions based on usage trends.

1.2.2.project excludes

Project Excludes
Physical Smart Fridge Development: The project focuses on software solutions and assumes the use of existing IoT-enabled fridges.
Automated Online Grocery Ordering: While the system provides shopping recommendations, it does not support direct purchases from online stores.
Voice Assistant Integration: The system does not include integration with Alexa, Google Assistant, or other voice-controlled devices in this phase.
Complex AI-Based Meal Planning: While the system suggests recipes based on available ingredients, it does not provide detailed meal plans or shopping lists.
Multi-User Inventory Tracking: The system is designed for individual use and does not yet support multiuser accounts with shared inventory tracking.
Remote Fridge Control: Users cannot control fridge settings (e.g., temperature adjustments) from the application.

1.3 Assumptions

Assumptions
The system assumes IoT sensors and computer vision will effectively track inventory in real-time.
It is assumed that users will occasionally need to manually verify and adjust inventory.
The recommendation system assumes continuous learning and improvement based on user interactions.
The system assumes development as a mobile or web-based application for broad accessibility.

1.4 Constraints

Constraints
The project does not include physical smart fridge development and relies on existing IoT-enabled fridges.
The system does not support automated online grocery ordering but can recommend needed items.
Voice assistant integration is not included in this phase but may be considered for future enhancements.

Section 2. Project Start-Up

2.1 Project Life Cycle

The Smart Fridge Management System project follows the System Development Life Cycle (SDLC) methodology, which provides a structured and systematic process to guide the development from initial planning to deployment and operation. The SDLC consists of four major phases:

Phase 1: Systems Planning and Selection

This phase begins with identifying the need for a new or enhanced system, such as improving food management and reducing waste through a smart fridge solution.

The team analyzes the organization's information needs, which may arise from existing limitations, desired functionalities, or the potential to utilize emerging technologies like AI and IoT.

A feasibility study is conducted to assess the viability of integrating IoT sensors for item tracking, and AI models for recipe recommendations and expiry detection.

A baseline project plan is developed, including the scope, objectives, timeframes, and required resources. This plan guides the development process and customizes the SDLC to fit the project's specific needs.

Phase 2: Systems Analysis

In this phase, the current methods of food storage and management are examined, and alternative smart system solutions are proposed.

Detailed requirements are gathered from potential users (e.g., families, nutritionists), including functional needs like expiration alerts and non-functional ones like system reliability and userfriendliness.

These requirements are analyzed, structured, and optimized to eliminate redundancies.

Several initial system designs are drafted, such as using computer vision for item recognition or using cloud-based databases.

A comparison of these alternatives is made to select the most effective and feasible solution within the project's constraints (cost, time, technical resources).

The output is a clear recommendation for the most suitable system architecture and features.

Phase 3: Systems Design

In this phase, the chosen solution is translated into both logical and physical system designs.

The logical design includes data flows, user interactions, and the definition of components like the AI decision engine and IoT communication logic, without tying it to any specific platform.

In physical design, technical specifications are developed for implementing the UI screens, report generation, sensor integration (e.g., RFID or weight sensors), cloud connectivity, and back-end services.

The team also designs the database schema for storing food item metadata, expiration dates, and user preferences.

Phase 4: Systems Implementation and Operation

This phase includes turning system specifications into a fully functioning solution.

During implementation, the team writes the code, trains the AI models, integrates hardware components, and builds the mobile/web application.

Comprehensive testing is performed at various levels: unit testing for components, system testing for overall functionality, and user acceptance testing to gather feedback.

The system is then deployed—application software is installed, IoT devices are connected, and users are onboarded.

Training materials and documentation are provided to ensure users can interact smoothly with the system.

The system enters the operational stage, where it becomes part of the user's routine.

Maintenance and updates are performed regularly to fix bugs, add new features, and ensure long-term stability.

2.2 Methods, Tools, and Techniques

Development Methodology

*The project follows an **Agile Development Methodology**, allowing for iterative development and continuous feedback.*

Standards, Policies, and Procedures

- **Coding Standards:** PEP 8 (Python), ESLint (JavaScript)
- **Security Policies:** Secure authentication and encrypted data storage
- **Data Privacy Compliance:** GDPR compliance for user data protection
- **Development Workflow:** CI/CD pipeline using GitHub Actions

Programming Languages

- **Python** (Backend, AI/ML processing)

- **JavaScript (React.js)** (Frontend development)
- **SQL** (Database management with PostgreSQL)

Reusable Code Repositories

- Open-source AI models for image recognition
- IoT libraries for sensor data processing
- Existing barcode scanning APIs

Tools & Technologies

- **Backend Framework:** Flask (Python-based web framework)
- **Frontend Framework:** React.js for UI development
- **Database Management:** PostgreSQL for structured data storage
- **IoT Technologies:** Raspberry Pi, weight sensors, barcode scanners
- **Version Control:** GitHub for collaborative development
- **Cloud Services:** AWS / Firebase for real-time data synchronization

Techniques Used

- **Computer Vision** for food recognition
- **Machine Learning** for recipe recommendations
- **IoT-based Data Collection and Analysis**

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2.3 Estimation Methods and Estimates

Estimation Methods and Estimates	
Description	[<u>Best</u> / <u>Most Likely</u> / <u>Worst</u>]
Effort in <u>person-months</u> or <u>person-hours</u>	[8,12,16] (months)

Schedule in calendar <u>months</u>	[9,13,18] (months)
Budget in dollars	[\$15,000, \$23,000, \$30,000]
Level of Uncertainty	[low, medium, high]

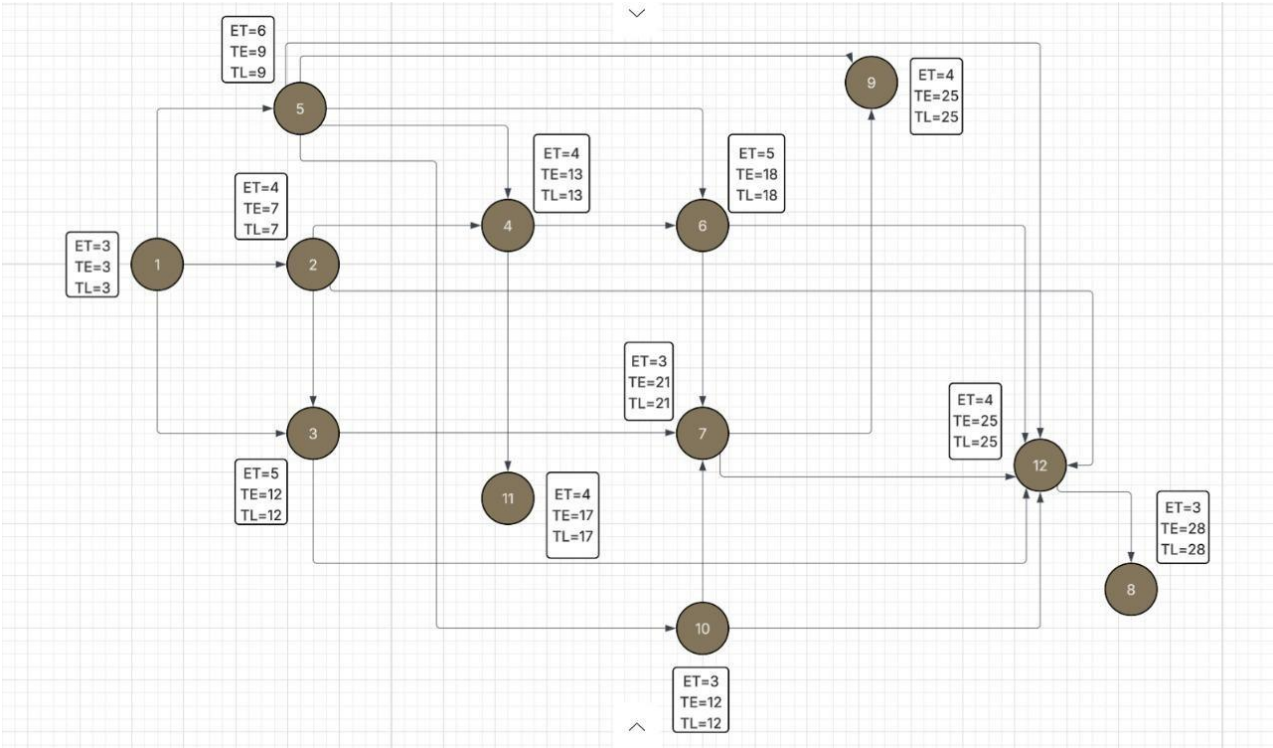
2.4 PERT:

Tasks	Optimistic time(o)	Realistic Time (r)	Pessimistic time(p)	Expected time(ET)
1-plan& Requirement	1	3	5	3
2-UI/UX Design	2	4	6	4
3-computer vision integration	3	5	7	5
4-smart notification system	2	4	6	4
5-Database Setup	2	4	6	6
6- Data Analysis & Recipe Recommendations	3	5	7	5
7-System Testing	1	3	5	3
8- Final App Launch	1	3	5	3
9- Hardware Integration	2	4	6	4

10- User Authentication & Account Management	1	3	5	3
11- Notifications & Alerts System	2	4	6	4
12- Final Integration & Debugging	2	4	6	4

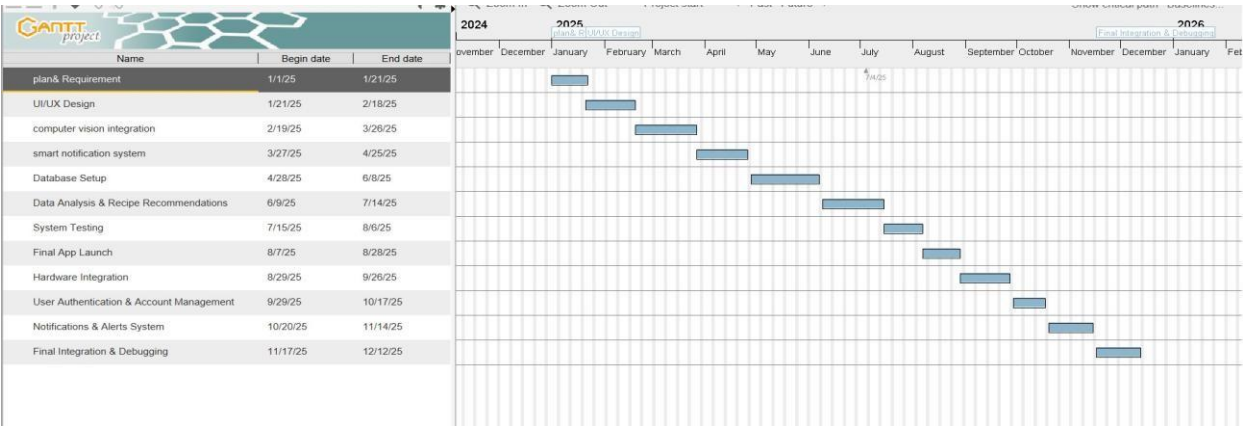
Tasks	Dependency
1-plan& Requirement	---
2-UI/UX Design	1
3-computer vision integration	1,2
4-smart notification system	2,5
5-Database Setup	1
6- Data Analysis & Recipe Recommendations	5,4
7-System Testing	3,10,6
8- Final App Launch	12
9- Hardware Integration	5,7
10- User Authentication & Account Management	5
11- Notifications & Alerts System	4
12- Final Integration & Debugging	7,2,5,10,3,6

2.5 Network Diagram



Tasks	Expected time(ET)	Earliest Time (TE)	Latest Time (TL)
1-plan& Requirement	3	3	3
2-UI/UX Design	4	7	7
3-computer vision integration	5	12	12
4-smart notification system	4	13	13
5-Database Setup	6	9	9
6- Data Analysis & Recipe Recommendations	5	18	18
7-System Testing	3	21	21
8- Final App Launch	3	28	28
9- Hardware Integration	4	25	25
10- User Authentication & Account Management	3	12	12
11- Notifications & Alerts System	4	17	17
12- Final Integration & Debugging	4	25	25

2.6 gantt chart



2.7 Resource Allocation

Resource	Total	Skill Set Requirements	Timeframe
Project Manager	1	Strong project management, leadership, and communication skills	Start: Month 1 Finish:
Software Developer	3	Proficient in C++, Python, IoT integration, and backend development	Start: Month 3– Finish: Month 8
Electronics Engineer (IoT Specialist)	1	Expertise in IoT, hardware design, and sensor integration	Start: Month 3 – Finish: Month 6
UI/UX Designer	1	Experience in UI/UX design and user interaction research	Start: Month 2 – Finish: Month 3
Month 9			
AI Specialist	1	Proficient in machine learning, data analysis, and recommendation systems	Start: Month 4 – Finish: Month 6
Software Tester (External)	1	Experience in unit testing, integration testing, and usability testing	Start: Month 6 – Finish: Month 8
Testing Equipment	3 Sets	Tools for debugging software and validating hardware functionality	Start: Month 6 – Finish: Month 8
Cloud Engineer			Start: Month 6 –

Facilities	1	Expertise in cloud services, backend infrastructure, and data synchronization	Finish: Month 8
	1	Space for development, testing, and collaboration	Start: Month 1 – Finish: Month 9

2.8 Budget Allocation

Key Budget Category	Budget Amount	Time Period
Personnel Costs (Salaries for team)	\$7,000	Throughout project duration
IoT Hardware & Sensors	\$5,000	Month 3
Software Development Tools		Month 3
Testing and Quality Assurance	\$2,500	Month 6
Cloud Services & Hosting	\$2,500	Month 4 - Month 9
Unexpected Expenses		
TOTAL	\$2,000	Throughout the project duration
	\$22,000	Within defined budget range

Note: The development phase (Months 3–8) involves significant implementation and integration tasks, requiring a team of three software developers to work concurrently.

2.9 Feasibility Study Analysis

Benefit per year =60,000

Recurring costs per year=40,000

Discount rate 10%

one time cost =22,000

Metric	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Net Economic Benefit	0	60,000	60,000	60,000	60,000	60,000
Discount Rate	1.000	0.909	0.826	0.751	0.683	0.621
PV of Benefits	0	54,540	49,560	45,060	40,980	37,170
NPV of All Benefits	0	54,540	104,100	149,160	190,140	227,310
Net Economic Cost	22,000	40,000	40,000	40,000	40,000	40,000
Discount Rate	1.000	0.909	0.826	0.751	0.683	0.621
PV of Cost	0	36,360	33,040	30,040	27,320	24,840
NPV of All Cost	22,000	58,360	91,400	121,440	148,760	173,600
Yearly NPV Cash Flow	-22,000	-3,820	12,700	27,720	41,380	53,710
Yearly PV Cash Flow	-22,000	18,180	16,520	15,020	13,660	12,330
Overall NPV						53,710

Rol= overall/ NPV(cost)

=53,710/173,600=0.097

Break-even analysis:

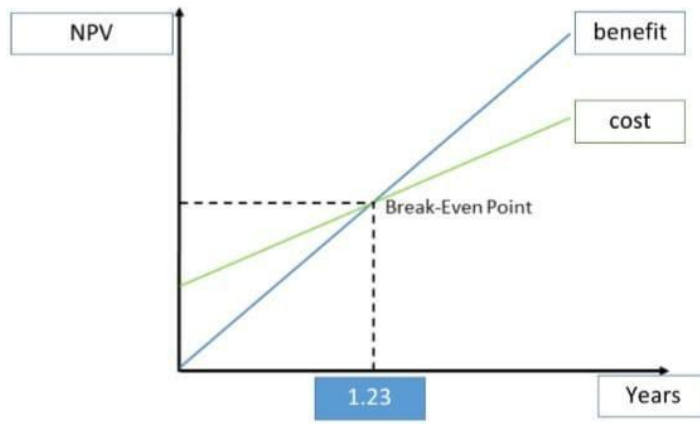
Break-even point =

(Yearly PV Cash Flow – Yearly Overall NPV Cash Flow) / Yearly PV Cash Flow

Break-even point =

(16,520 – 12,700) / 16,520 = 0.23

Actual Break-even occurred at year 1.23



Section 3. Risk Management

Based on project-specific methods, describe how risks will be analyzed to establish the project exposure for each risk and to determine which risks are the most important ones to address.

Risk Description	Probability	Impact	Strategy
<i>Cost Estimates Unrealistic</i>	Low	High	<i>Included in project plan, subject to amendment as new details regarding project scope are revealed</i>
<i>Time Estimates Unrealistic</i>	Medium	High	<i>Regular progress reviews, agile development approach to adapt to changes, and buffer time allocation in the schedule.</i>
<i>Team Size</i>	Medium	Medium	<i>Clearly define roles, consider outsourcing specific tasks if needed, and maintain flexible resource allocation.</i>
<i>Project Scope Creep</i>	High	High	<i>Defined in project plan, reviewed by Project Manager and Steering Committee to prevent scope creep</i>
<i>Team Members Unknowledgeable of Business</i>	Medium	High	Provide necessary training sessions on IoT, AI, and food safety. Ensure team collaboration with domain experts.

Available documentation	Low	Medium	Maintain up-to-date documentation, use version control tools, and establish a structured documentation process.
Narrow Knowledge Level of Users	Medium	High	Design user-friendly interfaces, conduct usability testing, and provide onboarding tutorials for users.

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Section 4. Appendices

4.1 Required Deliverables

The following deliverables are attached as part of this document to provide additional details about the project:

1. **Project Proposal** – Overview of the Smart Refrigerator system, objectives, and scope.
2. **System Architecture Diagram** – High-level architecture showcasing IoT integration, AI processing, and database management.
3. **Technical Stack Overview** – List of programming languages, frameworks, and tools used in the project.
4. **Use Case Scenarios** – Example scenarios demonstrating how users interact with the smart refrigerator.
5. **Risk Management Plan** – Detailed risk assessment and mitigation strategies.
6. **Project Schedule** – Timeline of milestones and expected completion dates.
7. **Budget Allocation** – Breakdown of estimated costs and financial planning.
8. **Testing and Quality Assurance Plan** – Strategy for software and hardware testing, including unit, integration, and usability testing.

4.2 Additional Resources

- **References & Research Papers** – Supporting materials related to IoT, AI in smart appliances, and sustainable food management.
- **Team Contact Information** – List of key team members and their roles for further inquiries.

Questionnaire

For whom: Regular Users / Home Users
Date: 5/7/2025

Questionnaire Questions

1. *What is the main purpose of using a smart fridge application in your daily life?*
2. *What are the most common problems you face when managing food in your fridge?*
3. *Do you think traditional fridges are enough to help manage food effectively? Why or why not?*
4. *Would it be helpful if the system could suggest recipes based on your available ingredients? How?*
5. *What types of information do you think are useful to manage your fridge better?*
(e.g., expiry dates, item quantity, food category)
6. *Should the system send alerts automatically when a product is about to expire or runs out?*
7. *How would you prefer to receive alerts?*
 - o Mobile Notification*
 - o Email*
 - o App Dashboard*
 - o Other*
8. *Should the system display a history of items and their expiry dates?*
9. *What platform would you prefer the system to run on?*
 - o Web*
 - o Mobile App*
 - o Desktop*
 - o All of the above*
10. *Are there any features or tools you'd like us to include in the system to make managing your fridge easier?*

Analysis

1. *What platform should the system work on?*
 2. *Should the system send alerts automatically?*
 3. *What alert methods do users prefer?*
 4. *What types of data are most useful for managing fridge contents?*
 5. *Would users like to see item history and expiry tracking?*
-

►The requirements that were obtained from this questionnaire are:

Functional Requirements:

1. The system should allow users to manage fridge contents in real time.
 2. It should detect items that are about to expire or run out.
 3. It should send alerts automatically when needed.
 4. Alerts should be sent to the user via:
 - Mobile Notification
 - Email
 - App Dashboard
 5. The system should show history of items and expiry dates.
 6. It should work on multiple platforms: Web – Mobile – Desktop.
 7. It should store a log of all alerts and fridge activity.
 8. Users should be able to suggest or request new features based on their needs.
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Non-Functional Requirements:

1. The system should work fast with no delays.
2. It should be easy to use and user-friendly for home users.
3. It should be secure and protect user data.
4. It should work 24/7 without crashing.
5. It should support different devices (mobile phones, tablets, computers).
6. It should be easy to update with new features or bug fixes.